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VACATION MINI PROJECT REPORT

BUILDCRAFT — PC BUILD RECOMMENDER

A browser-based PC build recommendation system for the Indian market

Submitted By	DARSHAN
Roll Number	24BCA072
Department	Bachelor of Computer Applications (BCA)
Semester	4th Semester
Academic Year	2026

CERTIFICATE

This is to certify that the mini project entitled

BUILDCRAFT — PC BUILD RECOMMENDER

has been successfully developed and submitted by **DARSHAN** (Roll No. **24BCA072**), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru.

This mini project was developed independently as part of self-learning during the vacation period. The work is original and has not been submitted elsewhere for any academic examination or award.

St. Claret College (Autonomous), Bengaluru

DECLARATION

I, **DARSHAN**, Roll No. **24BCA072**, 4th Semester, BCA, St. Claret College (Autonomous), Bengaluru, declare that the mini project **"BUILDCRAFT — PC BUILD RECOMMENDER"** is a record of original work done by me during the vacation period.

I declare that:

- This project was carried out independently, without any formal institutional guidance.
- The work is genuine and has not been copied from any source without acknowledgement.
- This project has not been submitted for any degree or academic title elsewhere.
- All references have been duly cited and acknowledged.
- The conclusions drawn are based entirely on my own understanding and research.

Place: Bengaluru

Date: July 2026

Signature of the Student

DARSHAN

Roll No: 24BCA072 | 4th Semester, BCA

ACKNOWLEDGEMENT

I thank all those who encouraged and supported me throughout the development of this project.

I am grateful to the open-source developer community, hardware review publications such as Tom’s Hardware, AnandTech, and GadgetsToUse, and the developer resources available through MDN Web Docs, Stack Overflow, and YouTube, which guided my understanding of component pricing logic, GST calculation, budget allocation algorithms, and JavaScript-based recommendation engine design.

I thank my family for their constant support and encouragement throughout the vacation period.

Building a tool that intelligently recommends custom PC configurations across use cases and budget ranges — with live INR pricing, 18% GST breakdown, and three curated build tiers — was both a technically demanding and deeply satisfying project that significantly broadened my understanding of front-end application design.

DARSHAN

Roll No: 24BCA072 | 4th Semester, BCA

St. Claret College (Autonomous), Bengaluru

ABSTRACT

Selecting the right PC components is a complex, high-stakes decision for students, gamers, content creators, and professionals alike. With hundreds of components across multiple generations, brands, and price points available in the Indian market, first-time builders frequently make suboptimal choices that result in budget overruns, performance bottlenecks, or compatibility issues. Existing tools such as PCPartPicker are international-focused and do not provide India-specific INR pricing with GST breakdown.

BuildCraft is a browser-based PC build recommendation engine tailored specifically for the Indian market. It guides users through a four-step process — use case selection, budget setting, preference configuration, and build generation — to produce a complete, invoice-ready PC component list with individual prices, GST-inclusive totals, and tier classification. Three curated preset builds (Entry, Mid-Range, High-End) are also provided for users who want a quick starting point.

Key Features:

- Use case selector: Gaming, Video/Photo Editing, Programming, Office Use, 3D Rendering, Streaming.
- Target resolution selector: 1080p, 1440p, 4K, or Not Applicable.
- Budget slider: 20,000 to 3,00,000 INR with real-time allocation.
- Optimise For selector: Balanced, CPU-heavy, GPU-heavy, Storage & Throughput, Longevity.
- CPU brand preference (Intel / AMD Ryzen) and GPU brand preference (NVIDIA / AMD Radeon).
- Form factor (ATX / Micro-ATX / Mini-ITX) and OS selection (Windows 11 Home/Pro / Linux).
- Optional includes: Monitor, Peripherals, UPS/Surge Protection, RGB lighting.
- Build Summary panel: Estimated Total with Base + 18% GST breakdown and per-component pricing.
- Save Build and Load Saved functionality using localStorage.
- Three curated tier presets: The Starter Rig (55k–80k), The Creator's Desk (1.2L–1.75L), The Workstation (2.5L–3.2L).
- 500+ components tracked across the Indian market price range 12k– 3L.

Technologies Used:

HTML5, CSS3, Vanilla JavaScript, localStorage API. Deployed on Netlify.

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INTRODUCTION

1.1 Background

The personal computer market in India has grown significantly over the past decade, driven by increased internet penetration, the rise of gaming culture, remote work, and content creation. However, building a custom PC remains a daunting task for most users due to the overwhelming number of component options, compatibility considerations, and rapidly fluctuating market prices.

A dedicated India-specific PC recommendation tool that understands local pricing with GST implications, respects INR budget constraints, and tailors recommendations to specific use cases — gaming, video editing, programming, or general office use — represents a significant practical need in the Indian market.

1.2 About the Project

BuildCraft is a four-step PC build recommendation web application designed specifically for the Indian market. It takes the user's use case, budget, brand preferences, form factor, and optional peripherals as inputs and generates a complete, component-by-component build list with individual INR prices and GST-inclusive totals. Three hand-curated preset builds (Entry Tier, Mid-Range Tier, High-End Tier) serve as quick-start templates.

Live at: **https://build-craft-pc.netlify.app/** — deployed on Netlify.

1.3 Scope

- India-specific PC component recommendations with INR pricing and 18% GST breakdown.
- Six use cases and four resolution targets driving component selection logic.
- Budget range 20,000 to 3,00,000 with intelligent allocation across components.
- CPU and GPU brand preference, form factor, and OS selection.
- Optional peripheral and UPS inclusion in the build budget.
- Three curated tier presets from Entry to High-End.
- Save and Load build functionality via localStorage.

OBJECTIVES OF THE PROJECT

The following objectives guided the design and development of BuildCraft:

1. To build an India-specific PC recommendation engine: Design a component selection algorithm that operates within an INR budget, accounts for 18% GST, and selects components based on Indian market availability and pricing.

2. To support diverse use cases: Accommodate six distinct computing use cases — Gaming, Video/Photo Editing, Programming, Office Use, 3D Rendering, and Streaming — each driving a different optimal component balance.

3. To implement intelligent budget allocation: Distribute the user's total budget intelligently across CPU, GPU, RAM, Storage, Motherboard, PSU, Case, Cooler, OS, and optional peripherals based on use case priority.

4. To provide transparent pricing with GST breakdown: Display each component's price individually and show the estimated total as Base Price + 18% GST, matching the invoice format used by Indian retailers.

5. To offer curated tier presets: Provide three hand-curated build templates — The Starter Rig, The Creator's Desk, and The Workstation — so users can start from a proven configuration.

6. To enable build persistence: Allow users to save their generated build to localStorage and reload it in a future session, preserving their configuration without any server dependency.

7. To deliver a premium editorial user experience: Design the interface with a dark, serif-forward, editorial aesthetic that communicates quality and reflects the premium nature of PC building as a craft.

EXISTING SYSTEM

3.1 Current PC Building Guidance Tools

Users currently rely on the following resources when planning a PC build, each with notable limitations for the Indian market:

a) PCPartPicker

PCPartPicker is the most popular PC build tool globally. However, it primarily serves the US and European markets with USD pricing. Its Indian edition exists but has limited retailer integration, does not show GST breakdowns, and lacks use-case-specific recommendation logic.

b) YouTube and Blog Guides

Many users rely on YouTube build guides or blog posts for recommendations. These are time-sensitive — component prices change frequently — and do not adapt to the user's specific budget, use case, or brand preferences.

c) Social Media and Forums

Reddit communities (r/IndianGaming, r/buildapc) provide personalised advice but require the user to post queries, wait for responses, and evaluate conflicting opinions. The process is slow and inconsistent.

d) Retailer Websites

Amazon India and Flipkart list components but provide no build compatibility checking, budget allocation logic, or use-case-based filtering. Users must manually research each component.

3.2 Limitations Summary

- No free India-specific tool provides use-case-driven PC build recommendations.
- Existing tools lack INR pricing with GST breakdown in a single interface.
- Manual research across YouTube, forums, and retailer sites is time-consuming and inconsistent.
- No tool intelligently allocates a given INR budget across all required components.
- No curated, tier-based build templates are available with verified Indian market prices.

PROPOSED SYSTEM

4.1 Overview

BuildCraft proposes a fully browser-based, India-specific PC build recommendation tool that accepts user inputs across use case, budget, and preferences, then generates a complete component list with INR pricing and GST-inclusive totals — all without any server or database dependency.

4.2 Advantages

- **India-First Design:** All pricing in INR with 18% GST shown explicitly. Component selections reflect Indian market availability across the 12k– 3L range.
- **Use-Case Intelligence:** Six distinct use cases drive different component priority weightings, ensuring the recommended build is optimised for the user's actual workflow.
- **Intelligent Budget Allocation:** The recommendation engine distributes the budget proportionally based on use case priorities, not a fixed template.
- **Transparent Pricing:** Per-component and total prices shown with GST breakdown, ready to verify against retailer invoices.
- **Tier Presets:** Three curated builds let users start from a proven configuration without going through the configurator.
- **Build Persistence:** Save and Load functionality lets users return to their build across sessions without losing configuration.
- **Zero Friction:** No login, no account, no server. Open the URL and get a build recommendation immediately.

4.3 Architecture

BuildCraft is a single-page application with three navigable sections: How It Works, Configurator, and Recommendations (Presets). The Configurator section contains a form panel and a live Build Summary panel. A JavaScript recommendation engine reads the form inputs and maps them to a component selection matrix, applying budget allocation ratios per use case. All generated build data is serialised to localStorage on Save.

MODULES AND FEATURES

BuildCraft comprises four primary sections accessible via the top navigation bar.

5.1 Hero / Landing Section

The landing section establishes BuildCraft's editorial identity with a dark near-black background and a large serif italic headline: "Your *ideal build,* priced in rupees." The tagline below explains the value proposition clearly. A 'Start Configuring' button with a right arrow links directly to the configurator.

Three statistics at the bottom of the hero section reinforce credibility: **500+** Components Tracked, **12k– 3L** Price Range Covered, and **3 tiers** (Budget · Mid · High).

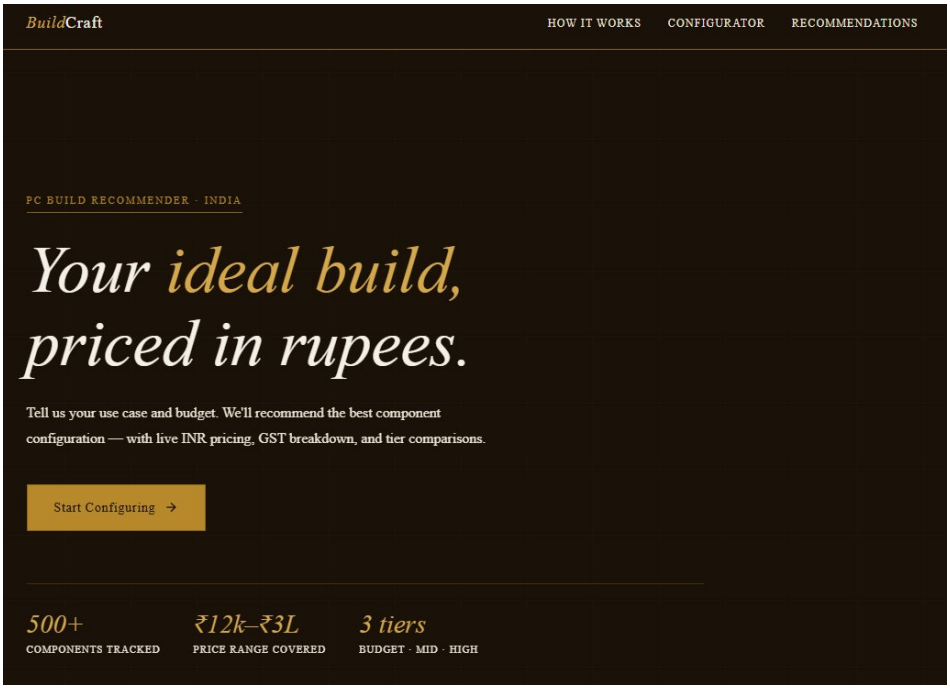


Figure 5.1 – BuildCraft Hero Section with Tagline and Statistics

5.2 How It Works Section

The 'How It Works' section presents the four-step build process in a clean card grid on a warm cream background, contrasting with the dark hero section:

- **Step 01 — Choose your use case:** Gaming, video editing, office work, or general use — the workflow determines the optimal component balance.
- **Step 02 — Set your budget:** Drag the slider to the total budget in INR. The engine allocates spend intelligently across components.

• **Step 03 — Configure preferences:** Brand preferences, form factor, and upgrade path priority fine-tune the recommendation.

• **Step 04 — Get your build:** Receive a full component list with individual and total pricing, GST included, ready to share or save.

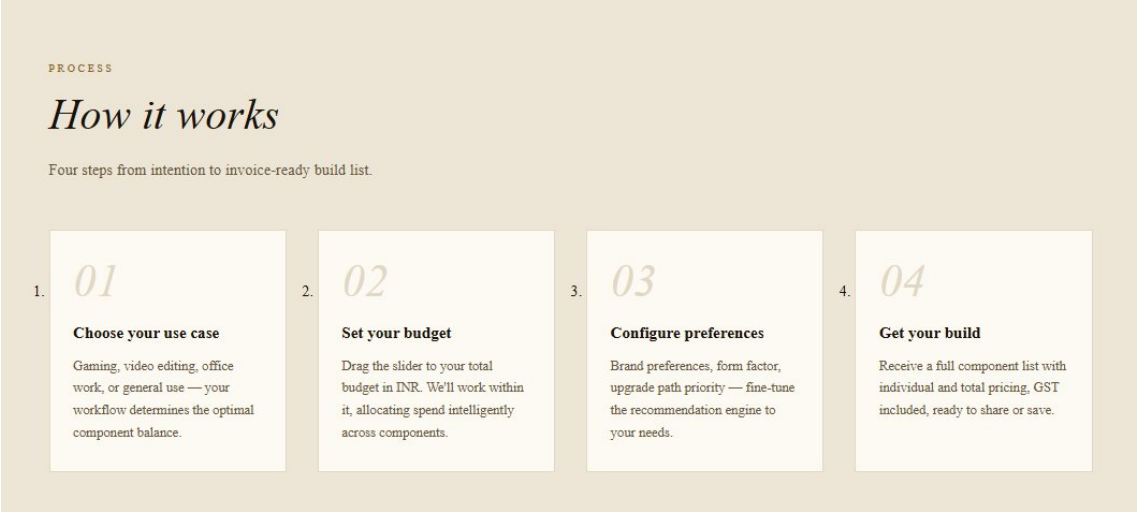


Figure 5.2 – How It Works Section with Four-Step Process Cards

5.3 Curated Builds by Tier (Presets)

The Presets section presents three hand-curated PC build templates on the cream background, each representing a distinct tier of the Indian PC market:

Tier	Name	Price Range	CPU	GPU	RAM	Storage
Entry	The Starter Rig	55,000– 80,000	AMD Ryzen 5 5600	RX 6600 8GB	16GB DDR4 3200	500GB NVMe
Mid	The Creator's Desk	1,20,000– 1,75,000	Intel i5-13600K	RTX 4060 Ti 8GB	32GB DDR5 5600	1TB NVMe Gen4
High	The Workstation	2,50,000– 3,20,000	Intel i9-13900K	RTX 4080 Super	64GB DDR5 6000	2TB NVMe Gen4

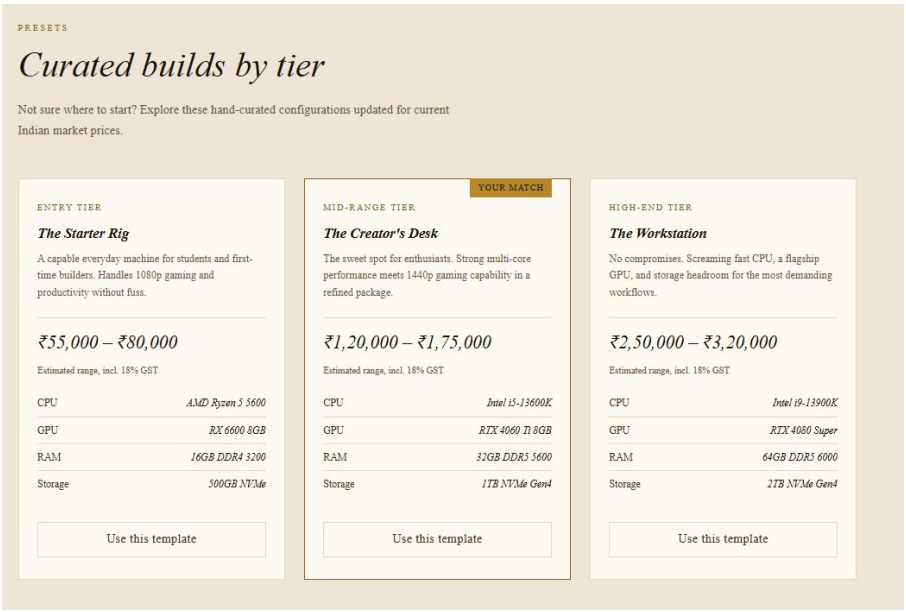


Figure 5.3 – Curated Builds by Tier with Three Preset Templates

5.4 Configurator Module

The Configurator is the core functional module of BuildCraft. It is divided into a left-side form panel and a right-side live Build Summary panel.

Form Panel — Use Case & Performance:

- **Primary Use Case:** Dropdown with six options: Gaming, Video/Photo Editing, Programming/Development, Office & General Use, 3D Rendering/CAD, Streaming/Content Creation.
- **Target Resolution:** 1080p (Full HD), 1440p (QHD), 4K (UHD), or Not Applicable.
- **Optimise For:** Balanced performance, CPU-heavy tasks, GPU-heavy tasks, Storage & throughput, or Longevity & upgradability.

Budget (INR, incl. GST):

A range slider from 20,000 to 3,00,000 with the current value displayed in large text below the slider. The recommendation engine uses this as the hard ceiling for the total build cost including GST.

Preferences:

- **CPU Brand Preference:** No preference, Intel, or AMD Ryzen.
- **GPU Brand Preference:** No preference, NVIDIA GeForce, or AMD Radeon.
- **Form Factor:** ATX (Full/Mid Tower), Micro-ATX, or Mini-ITX (Compact).
- **Operating System:** Windows 11 Home, Windows 11 Pro, or Linux (no OS cost).

Include in Build Toggles:

- **Monitor:** Include a display in the budget allocation.
- **Peripherals:** Keyboard, mouse, and headset.
- **UPS / Surge Protection:** Add a power backup unit to the build.
- **RGB Lighting:** Prefer components with RGB aesthetics.

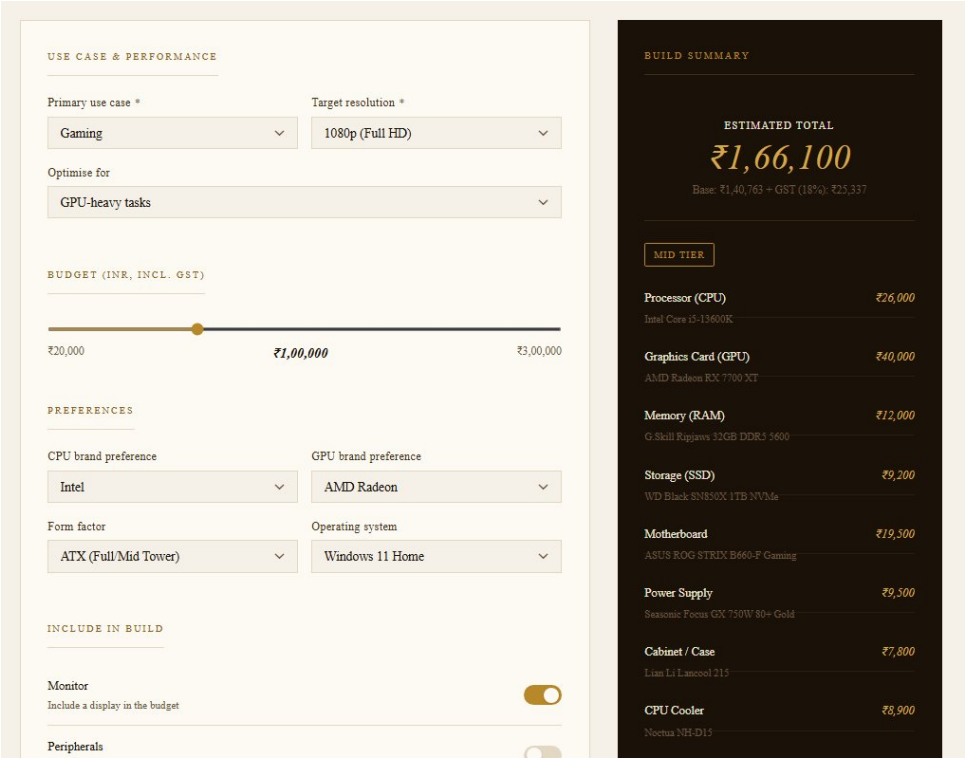


Figure 5.4 – Configurator Form Panel with Build Summary Panel

5.5 Build Summary Panel

The Build Summary panel on the right side of the configurator displays the generated component list in real time after clicking ‘Generate Recommendation’. The panel shows:

- **Estimated Total:** Displayed prominently in gold serif type (e.g., 1,66,100) with the Base price + 18% GST breakdown shown below.
- **Tier Badge:** Classifies the build as Entry, Mid Tier, or High-End based on the total cost.
- **Component List:** Each component (Processor, Graphics Card, Memory, Storage, Motherboard, Power Supply, Cabinet, CPU Cooler, Operating System, Monitor, UPS) is listed with its specific model name and INR price.
- **Save Build:** Serialises the current build configuration to localStorage.
- **Load Saved:** Restores the previously saved build from localStorage.

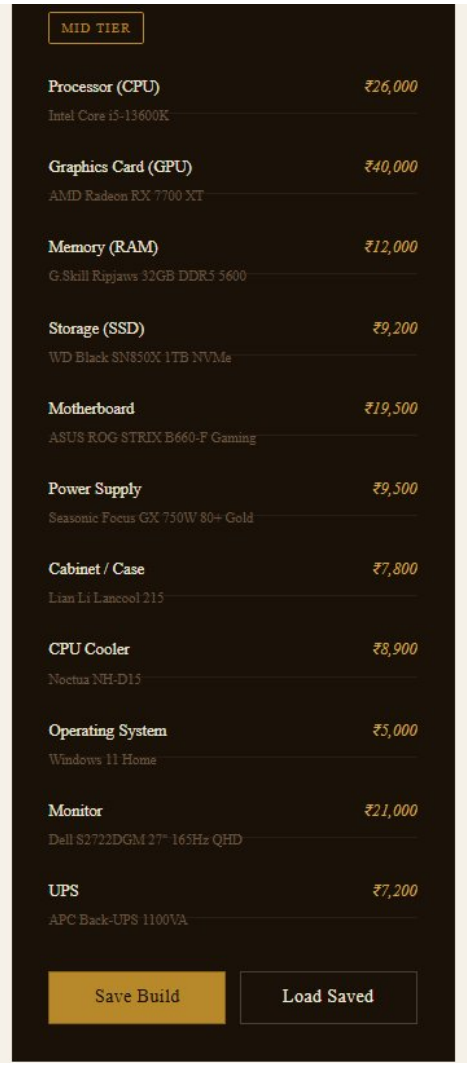


Figure 5.5 – Build Summary Panel Showing Mid Tier Component List with Prices

TECHNOLOGIES USED

BuildCraft was developed exclusively using native front-end web technologies. No back-end server, database, or external API is used.

6.1 HTML5

HTML5 provides the structural foundation of BuildCraft's three-section layout. Semantic elements organise the hero, how-it-works, configurator, and presets sections. Standard form controls — select dropdowns, range sliders, toggle checkboxes, and radio buttons — power the configurator input panel. Smooth-scroll anchor links connect the navigation bar to the corresponding sections.

6.2 CSS3

CSS3 delivers BuildCraft's distinctive dual-aesthetic design: a dark near-black background with gold serif typography for the hero section, contrasting with a warm cream/parchment background for the configurator and presets sections. CSS Custom Properties manage the colour system. Flexbox and CSS Grid handle the responsive four-column how-it-works card layout and the two-column configurator/summary split. CSS transitions animate the range slider thumb and toggle states.

6.3 JavaScript (ES6+, Vanilla)

All recommendation logic is implemented in Vanilla JavaScript. Key algorithms include: a use-case-to-component-weight mapping that determines budget allocation ratios (e.g., Gaming allocates a higher proportion to GPU), a budget tier classification function (Entry below 80k, Mid 80k– 1.8L, High above 1.8L), a component selection function that maps budget allocations to specific component models from a pre-defined catalogue, a GST calculation function that computes 18% tax on the base total, and build persistence functions using JSON serialisation to localStorage.

6.4 localStorage API

The Web Storage API's localStorage interface is used to implement the Save Build and Load Saved features. The complete build configuration — including use case, budget, preferences, and the generated component list — is serialised as a JSON object and stored in localStorage, enabling the user to return to their build in a future browser session.

6.5 Netlify

The application is deployed as a static site on Netlify, providing HTTPS, a custom subdomain, and global CDN distribution with no server-side infrastructure required.

#	Technology	Type	Purpose
1	HTML5	Markup	Structure, form controls, anchor navigation
2	CSS3	Styling	Dual-theme design, grid layout, transitions
3	JavaScript ES6+	Programming	Recommendation engine, budget allocation, GST
4	localStorage	Browser API	Save and Load build persistence
5	Netlify	Platform	Static hosting and CDN deployment

CONCLUSION

BuildCraft is a thoughtfully designed and technically accomplished mini project that addresses a genuine, underserved need in the Indian technology market. By building an intelligent PC recommendation engine that is specifically calibrated for INR pricing, 18% GST, and Indian market component availability — and packaging it in a premium, editorial interface that rivals commercial products in visual quality — the project demonstrates both technical skill and a clear sense of product purpose.

The recommendation engine's ability to intelligently allocate a given INR budget across eleven component categories based on use case priority weights is the project's most significant technical achievement. Implementing this logic in Vanilla JavaScript without any external library required careful design of data structures, allocation algorithms, and tier classification rules.

The dual-aesthetic design — dark and editorial for the hero section, warm cream and serif-forward for the configurator and presets — is deliberately crafted to reflect the premium, considered nature of PC building as a craft. This level of design intentionality distinguishes BuildCraft from a generic utility tool.

Completing this project independently during the vacation period has been a highly rewarding experience in applied front-end development. Researching Indian market component pricing across 500+ components, designing the allocation algorithm, building the configurator UI, and deploying a polished public product has provided comprehensive exposure to the full product development lifecycle.

In conclusion, BuildCraft is a polished, feature-complete, and publicly deployed PC build recommendation tool that fully achieves its stated objectives and stands as a strong demonstration of front-end development skill, India-specific product thinking, and self-directed technical initiative.

Live Project: https://build-craft-pc.netlify.app/